

An Analytic-Eigenfield Criterion for Injectivity in Abstract de Branges–Rovnyak Interpolation

Run fa_banach_001, agent agent_lane_11

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Source Question and Status

Ball, Bolotnikov, and ter Horst associate to admissible abstract interpolation data (P, T, E, N) a unitary colligation with characteristic function $\Sigma = (\Sigma_{ij})$. After proving sufficient conditions for the multiplication operator

$$M_{\Sigma_{12}} : \text{Hol}_{\Delta_*}(\mathbb{D}) \longrightarrow \text{Hol}_Y(\mathbb{D})$$

to be injective, they write: “The question of finding a condition that is both necessary and sufficient for injectivity of $M_{\Sigma_{12}}$ remains open” [1, p. 16].

Thus $M_{\Sigma_{12}} : \text{Hol}_{\Delta_*}(\mathbb{D}) \rightarrow \text{Hol}_Y(\mathbb{D})$ is injective. □

It is easy to see that not only the injectivity of $M_{\Sigma_{12}}$, but all the conclusions of Theorem 4.5 hold with the condition that T^* is injective replaced by the weaker condition (4.32). Although condition (4.32) is far from being necessary, it guarantees injectivity of $M_{\Sigma_{12}}$ for important particular cases:

- (1) T^* is injective (so $\text{Ker } T^* = \{0\}$),
- (2) T^* is nilpotent (so $\cap_{k \geq 1} \text{Ran}(T^*)^k = \{0\}$), and
- (3) $\dim \mathcal{X} < \infty$, or, more generally e.g., $T = \lambda I + K$ with $0 \neq \lambda \in \mathbb{C}$ and K compact (so $\mathcal{X} = \text{Ran}(T^*)^p \dot{+} \text{Ker}(T^*)^p$ once p is sufficiently large).

The question of finding a condition that is both necessary and sufficient for injectivity of $M_{\Sigma_{12}}$ remains open.

5. DESCRIPTION OF ALL SOLUTIONS OF THE PROBLEM $\mathbf{AIP}_{\mathcal{H}(K_S)}$

We now present the parametrization of the solution set to the problem $\mathbf{AIP}_{\mathcal{H}(K_S)}$. The proof relies on Theorem 3.3, Theorem 4.1 and Proposition 4.2. By Theorem

This packet gives such a condition directly in terms of the original operators P and T . It applies without a nondegeneracy assumption on P . Status: *candidate full solution, likely valid*, subject to human review.

Setup and the Source Criterion

Let $\mathcal{X}$ be the state Hilbert space, let $P \geq 0$, and put

$$\mathcal{X}_0 = \overline{\text{Ran } P^{1/2}}.$$

The restriction of $P^{1/2}$ to $\mathcal{X}_0$ is injective. The source's unitary colligation has state and input blocks A and B_2 ; in particular, $\|A\| \leq 1$ and $\|B_2\| \leq 1$.

Lemma 4.7 of [1] says that $M_{\Sigma_{12}}$ is noninjective if and only if there are nonzero $g_n \in \mathcal{X}_0$ such that

$$T^*P^{1/2}g_1 = 0, \quad (1)$$

$$T^*P^{1/2}g_{n+1} = P^{1/2}g_n \quad (n \geq 1), \quad (2)$$

$$\limsup_{n \rightarrow \infty} \|B_2^*g_{n+1}\|^{1/n} \leq 1. \quad (3)$$

The point of the result below is to remove the colligation-dependent block B_2 and package the entire chain as one holomorphic operator-pencil equation.

Proof Intuition

The chain in (1)–(2) is already the Taylor coefficient recursion for a generalized eigenvector field $G(z) = \sum_{n \geq 0} g_{n+1}z^n$. The only apparent mismatch is that the source controls $B_2^*g_{n+1}$ rather than g_{n+1} . In the converse proof of Lemma 4.7, however, the same coefficients satisfy

$$g_{n+1} = Ag_n + B_2B_2^*g_{n+1}.$$

Because A is a contraction, unit-disk Taylor growth of $B_2^*g_{n+1}$ is equivalent to unit-disk Taylor growth of g_{n+1} . This turns the source's sequence condition into a single intrinsic equation involving only P, T .

Necessary-and-Sufficient Condition

Theorem 1 (analytic-eigenfield criterion). *For any admissible data set considered in [1], the following are equivalent.*

1. $M_{\Sigma_{12}} : \text{Hol}_{\tilde{\Delta}^*}(\mathbb{D}) \rightarrow \text{Hol}(\mathbb{D})$ is injective.
2. The only norm-holomorphic function $G : \mathbb{D} \rightarrow \mathcal{X}_0$ satisfying

$$T^*P^{1/2}G(z) = zP^{1/2}G(z) \quad (z \in \mathbb{D}) \quad (4)$$

is $G \equiv 0$.

Thus (4) is a necessary-and-sufficient condition expressed solely in the original P, T data, including when P is degenerate.

Proof. We first isolate the growth equivalence missing from the source statement. Suppose g_n satisfies the algebraic relations (1)–(2). In the converse half of the proof of Lemma 4.7, set

$$x_0 = g_1, \quad x_n = g_{n+1} - Ag_n, \quad y_0 = 0, \quad y_n = -C_1g_n \quad (n \geq 1),$$

and $h_n = \mathbf{i}_*[x_n \ y_n]^\top$. The algebraic computation in that proof, before any growth assumption is used, gives

$$h_n = B_2^*g_{n+1}, \quad x_n = B_2h_n. \quad (5)$$

The second equality follows as well from the colligation identity $[x_n \ y_n]^\top = [B_2 \ D_{12}]^\top h_n$. Hence, for $n \geq 1$,

$$g_{n+1} = Ag_n + B_2 h_n. \quad (6)$$

If (3) holds, then for every $r > 1$ there is C_r such that $\|h_n\| \leq C_r r^n$. Iterating (6) and using $\|A\|, \|B_2\| \leq 1$ gives

$$\|g_{n+1}\| \leq \|g_1\| + C_r \sum_{j=1}^n r^j,$$

and therefore

$$\limsup_{n \rightarrow \infty} \|g_{n+1}\|^{1/n} \leq 1. \quad (7)$$

Conversely, (7) implies (3) immediately from $h_n = B_2^* g_{n+1}$. Thus the two growth conditions are equivalent.

Assume first that $M_{\Sigma_{12}}$ is not injective. Lemma 4.7 supplies a chain satisfying (1)–(3); by the preceding paragraph it also satisfies (7). Consequently

$$G(z) = \sum_{n=0}^{\infty} g_{n+1} z^n$$

defines a nonzero norm-holomorphic $\mathcal{X}_0$ -valued function on $\mathbb{D}$. Termwise application of the bounded operators $P^{1/2}$ and T^* , followed by (1)–(2), gives

$$T^* P^{1/2} G(z) = \sum_{n \geq 1} P^{1/2} g_n z^n = z P^{1/2} G(z).$$

Conversely, suppose a nonzero holomorphic G satisfies (4). Divide by its largest common initial power of z so that, without loss of generality, $G(0) \neq 0$; equation (4) is preserved. Write $G(z) = \sum_{n \geq 0} g_{n+1} z^n$. The Banach-valued Cauchy–Hadamard formula gives (7), while coefficient comparison in (4) gives (1)–(2). The growth equivalence gives (3). Lemma 4.7 now says that $M_{\Sigma_{12}}$ is not injective. This proves the equivalence. $\square$

Remark 1. If P is positive definite, define the bounded similar pencil operator $L_P = P^{-1/2} T^* P^{1/2}$. The criterion says that L_P has no nonzero holomorphic eigenvector field G on the whole unit disk with $L_P G(z) = z G(z)$. For degenerate P , formula (4) avoids the unbounded inverse of $P^{1/2}$ entirely.

Weighted Shifts: Sharpness of the Criterion

Corollary 1. Let $\mathcal{X} = \ell_2(\mathbb{N}_0)$, $P = I$, and

$$T e_n = w_n e_{n+1}, \quad 0 < w_n \leq 1.$$

Take $N = 0$ and $E = (I - T^* T)^{1/2}$, so the Stein identity holds. Then $M_{\Sigma_{12}}$ is noninjective if and only if

$$\liminf_{n \rightarrow \infty} (w_0 w_1 \cdots w_{n-1})^{1/n} = 1. \quad (8)$$

In particular, for $w_n = 1/(n+1)$ the paper’s sufficient condition (4.32) fails, but $M_{\Sigma_{12}}$ is injective.

Proof. Here $T^*e_0 = 0$ and $T^*e_{n+1} = w_n e_n$. If a nonzero field satisfies $T^*G(z) = zG(z)$, division by its initial zero makes $G(0) = ce_0$ with $c \neq 0$. Comparing Taylor coefficients shows that the coefficient of e_n in the n th Taylor coefficient is forced to equal

$$\frac{c}{w_0 w_1 \cdots w_{n-1}}.$$

Conversely, choosing precisely these forced vectors gives a formal solution. It is holomorphic on $\mathbb{D}$ exactly when their root growth is at most one, which, since every $w_n \leq 1$, is equivalent to (8). Theorem 1 proves the first assertion.

For $w_n = 1/(n+1)$, one has $e_0 = (T^*)^n(n!e_n)$ for every n , so $e_0 \in \text{Ker } T^* \cap \bigcap_{n \geq 1} \text{Ran}(T^*)^n$ and (4.32) fails. On the other hand, every prospective eigenfield has a coefficient of norm at least $|c|n!$, whose root growth is unbounded. No nonzero field is holomorphic on $\mathbb{D}$, so the multiplier is injective by Theorem 1. $\square$

Scope, Novelty Check, and Verification

The result answers the source’s injectivity question exactly. It is a structural criterion, not a claim of a finite algorithm for arbitrary infinite-dimensional data. Its content beyond Lemma 4.7 is the elimination of B_2 and the equivalent holomorphic pencil formulation; the weighted shift example shows that this retains cases missed by condition (4.32).

The run’s cheap indexes contained no answer. A bounded search through 17 August 2026 used the exact open sentence, the title and authors, and the phrases *Redheffer injectivity* Σ_{12} , *analytic core Redheffer transform*, and *de Branges–Rovnyak analytic core*. It found the source, surveys, and applications, but no later answer or this criterion.

The companion script checks the weighted-product formulas, the hyperrange witnesses, and the factorial growth for finite indices. These computations are consistency checks only; the proof above is analytic.

Human Review Recommendation

Check that identities (5) follow from the source’s converse construction before its growth hypothesis is used, and then verify the contraction estimate leading to (7). Those are the only new proof steps on which the full criterion depends.

References

- [1] J. A. Ball, V. Bolotnikov, and S. ter Horst, *Abstract interpolation in vector-valued de Branges–Rovnyak spaces*, Integral Equations Operator Theory **70** (2011), 227–263; arXiv:1804.08492.